

YOLOv11 based algorithm for abandoned luggage detection with dynamic radius estimation

Ivan Vrsalovic
Faculty of Engineering, University of Rijeka
Faculty of Informatics and Digital Technologies, University of Rijeka
Rijeka, Croatia
ivan.vrsalovic@uniri.hr

Marina Ivasic-Kos
Faculty of Informatics and Digital Technologies, University of Rijeka
Centre for Artificial Intelligence, University of Rijeka
Rijeka, Croatia
ORCID: 0000-0002-1940-5089

Miran Pobar
Faculty of Informatics and Digital Technologies, University of Rijeka
Centre for Artificial Intelligence, University of Rijeka
Rijeka, Croatia
ORCID: 0000-0001-5604-2128

Jonatan Lerga
Faculty of Engineering, University of Rijeka
Centre for Artificial Intelligence, University of Rijeka
Rijeka, Croatia
ORCID: 0000-0002-4058-8449

Abstract— We address the challenge of abandoned luggage detection by adapting a YOLOv11 model and validating it on the IITP20 dataset. Our methodology involves optimizing our pre-trained YOLOv11 model for the domain adaptation to specific characteristics of the IITP20 dataset. Our initial analysis revealed significant semantic inconsistencies within the dataset's annotations; a large portion of "abandoned" luggage instances were simply items in lockers or storage areas, which do not constitute security threats. To create a more realistic benchmark, we systematically corrected these inaccuracies. A comparative analysis reveals that our person and baggage detection model, along with the proposed algorithm, achieves state-of-the-art results, demonstrating exceptional performance in scenarios relevant to security surveillance. These findings underscore the robustness of our detection approach and emphasize the importance of high-quality, semantically accurate annotations for developing reliable abandoned object detection systems in real-world applications.

Keywords— Abandoned Luggage Detection, YOLOv11, Depth pro, Depth Estimation, Surveillance Systems

I. INTRODUCTION

Video surveillance systems have become an indispensable tool in ensuring public safety and property protection, playing a crucial role in detecting possible threats such as abandoned objects. The rapid and reliable identification of abandoned luggage, particularly in densely populated and dynamic environments like airports, train stations, and shopping centers, is vital for preventing potential security incidents. Despite significant advancements in the field of computer vision, the development of robust and accurate abandoned object detection systems continues to pose a considerable challenge due to scenario variability, lighting conditions, occlusions, and the diversity of the objects themselves.

In this paper, we focus on optimizing our fine-tuned YOLOv11 model, specialized for detecting people and luggage captured by surveillance cameras in public spaces, and validating its robustness to domain changes together with the abandoned luggage detection algorithm. The evaluation is based on the publicly available IITP20 dataset [1], which contains real-world scenes for testing abandoned object detection systems. However, our initial analysis of this dataset

revealed significant challenges in the accuracy and consistency of existing annotations, particularly in scenarios of medium and high complexity. We frequently observed instances where luggage was incorrectly labeled as "abandoned" despite being in a context suggesting temporary storage, such as luggage lockers or other designated areas for placing belongings, leading to detections that lack practical utility in surveillance systems. Thus, we performed the experiments on the corrected dataset.

The objectives of this work are fourfold:

- To detail the process of adapting and optimizing our YOLOv11 model, leveraging an existing pre-trained version through fine-tuning to demonstrate effective transfer learning, for efficient processing and detection on the IITP20 dataset,
- To propose a new methodology for recognizing abandoned luggage with dynamic radius estimation for person-baggage connection,
- To conduct a thorough evaluation performance of our detection model and abandoned luggage detection algorithm on IITP20 dataset,
- To compare the obtained results with existing methods.

The structure of the paper is as follows: Section 2 provides an overview of related work in the field of abandoned luggage detection, including traditional and modern approaches. Section 3 describes the methodology for fine-tuning YOLOv11 model for detecting people and luggage captured with surveillance cameras and its adaptation to the IITP20 dataset. Section 4 describes the abandoned luggage detection algorithm and its key steps and parameters. Section 5 provides experimental analysis and comparison with previously published results on the same dataset. Finally, Section 6 presents conclusions and future research directions.

II. RELATED WORK

Automated abandoned object detection in video surveillance requires precise localization and identification of luggage, followed by an accurate determination of abandonment. This domain has evolved from early computer

This research was partially supported by the UNIRI project SAR-DAG

vision techniques, such as background subtraction [2], [3], which proved fragile in dynamic and complex scenes, to advanced deep learning methodologies [4], [5], [14]. Contemporary object detection frameworks, notably those based on the You Only Look Once (YOLO) architecture (e.g., YOLOv8, YOLOv11) [7], now offer superior real-time inference and robust accuracy across varied and challenging surveillance environments [6], [8], [14]. Despite these advancements, inherent difficulties persist, including effective detection of small objects in footage where CNN down-sampling can lead to a loss of critical spatial details [8].

A fundamental challenge hindering the development of truly robust abandoned object detection systems is the acute scarcity of diverse and representative real-world data, particularly for rare events like actual luggage abandonment [9]. This data limitation hinders comprehensive training and validation, often resulting in models that perform adequately on common scenarios but struggle when confronted with infrequent yet critical situations. Consequently, the strategic creation of varied and challenging scenarios becomes indispensable for developing reliable detection algorithms. Most commonly, actor-based simulated scenarios are used to create luggage abandonment datasets, as they are effective in producing realistic and context-rich scenarios. However, due to logistical constraints, the most common datasets, such as PETS2006[10] and AVSS 2007[11], are small and have become difficult to obtain and are no longer available.

Recently, [1] introduced a significant contribution with the IITP20 dataset, a collection of 58 videos capturing numerous complex and challenging abandoned object scenarios, including group ownership, occlusions, object exchanges, and person re-entry, which are vital for thorough system evaluation. Alongside this dataset, they proposed an integrated detection framework employing a customized RetinaNet-based architecture for small object detection, combined with an algorithm for object association and owner identification. Their approach emphasizes spatial and temporal reasoning, using both distance-based and duration-based thresholds for abandonment through homographic transformations to measure real-world distances [1]. However, a limitation of relying on homography for spatial reasoning is its dependency on a predefined ground plane, which struggles to accurately measure distances or assess proximity for objects placed on elevated surfaces such as chairs or platforms, a common occurrence in real-world scenarios, as seen in the IITP20 demo. To overcome these constraints, in [18], depth estimation models were incorporated to aid with spatial reasoning and determine spatial relationships between people and luggage in the 3D scene.

In our previous work [14, 18], we explored various approaches for generating rare and challenging abandoned luggage scenarios, aiming to create comprehensive test sets for algorithm evaluation. Through detailed analysis, we determined that scene augmentation and careful video editing methods were the most effective for creating controlled and realistic test cases that simulate desired behaviors, providing a platform for precise testing and model optimization.

Our previous studies have not evaluated the robustness of the detection model and abandoned baggage recognition algorithm under domain shifts, so this paper aims to test this on the publicly available IITP20 dataset adapted for

abandoned luggage detection tasks and critically assess the applicability in real-world surveillance.

III. FINE-TUNED PERSON AND LUGGAGE DETECTION MODEL

A. YOLOv11 Model Fine-tuning and Training Environment

To optimize detecting luggage and persons performance in airport-like surveillance conditions, YOLOv11 was initially fine-tuned on the CCTV-Surveillance Dataset [14]. This custom dataset consists of 238 video segments, combining outdoor city promenade footage (Korzo, Croatia) with indoor airport scenes (Düsseldorf Airport, Germany) to represent a diverse public space environment (Fig. 1). It contains 2,676 person detections and 880 luggage detections. To enhance data variability, augmentation techniques such as horizontal flipping, rotation, saturation shifts, hue shifts, exposure adjustments, and noise addition were applied, expanding the dataset to 450 images.

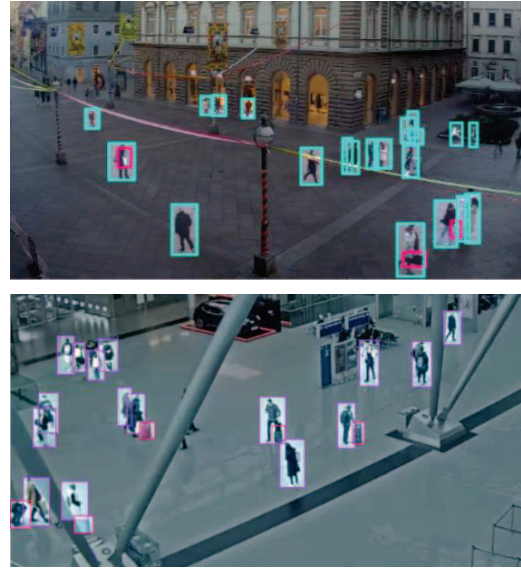


Fig. 1. Example images from CCTV Surveillance dataset (Korzo and airport) used for YOLOv11 model fine-tuning

The fine-tuning on the CCTV-Surveillance Dataset focused on “luggage” and “person” classes. Training employed a standard loss function with the Sigmoid Linear Unit (SiLU) activation function. A batch size of 16 and a momentum of 0.937 were used, with images uniformly resized to 640x640 pixels. Models were trained for up to 50 epochs or until convergence.

After this fine-tuning, all YOLOv11 versions significantly improved their performance compared to baselines trained on the COCO dataset, which achieved mAP@0.5 of 4% and recall of 22% (Table 1). The fine-tuned YOLOv11-m version was selected for our abandoned luggage algorithm due to its optimal balance of accuracy and computational efficiency, achieving a mAP@0.5 of 76% and recall of 80% on the CCTV-Surveillance test set. The full performance metrics for the baseline and fine-tuned YOLOv11 versions are detailed in Table 1. This initial fine-tuning establishes a strong baseline for subsequent transfer learning [15] to the IITP20 dataset.

Table 1 Comparison of default and fine-tuned YOLOv11 detection results on CCTV Surveillance test set

Metric	YOLOv11-s	YOLOv11-m	YOLOv11-l
Default YOLO version			
Precision	0.059	0.079	0.508
Recall	0.218	0.202	0.065
mAP@50	0.032	0.042	0.011
Versions fine-tuned on CCTV training set			
Precision	0.74	0.76	0.75
Recall	0.73	0.80	0.81
mAP@50	0.75	0.76	0.76

B. Model Adaptation for Domain Shift on the IITP20 Dataset

Building upon the initial fine-tuning on the CCTV-Surveillance Dataset, the next step involved adapting the selected YOLOv11-m model for the IITP20 dataset.

For the fine-tuning process, 100 unique images from the IITP20 dataset were selected (Fig. 2). This small subset was chosen to test the fast transfer of knowledge from the extensively pre-trained model.



Fig. 2. Object detection fine-tuning training example from the IITP20 dataset

The previously established training configurations were largely maintained, including the continued focus on “luggage” and “person” classes, batch size of 16, and the 640x640 pixel input image resolution. The model was trained for a determined number of epochs (150), or until performance convergence was observed (90 epochs in our case) on a dedicated validation split of this small IITP20 subset. Following this fine-tuning process on the IITP20 dataset, our YOLOv11-m model, marked YOLOv11-m (CCTV+IITP20) achieved remarkable performance across key metrics (Table 2), showing that for adaptation on a new environment or surveillance system, a small number of additional training images is sufficient.

The achieved object detection performance on the IITP20 dataset exceed the previously published results on that data set [1], which was 70% average precision with an unknown IoU, with all fine-tuned YOLOv11-m models, including the YOLOv11-m (CCTV) model, which was fine-tuned only on CCTV images and not on IITP20 set.

Table 2. Object detection results on IITP20 dataset

Metric	YOLOv11-m				ResNet (Amin et. al [1])
	Default	CCTV	IITP20	CCTV+IITP20	
Precision	0.52	0.76	0.94	0.96	/
Recall	0.19	0.80	0.94	0.93	/
mAP@50	0.29	0.76	0.93	0.93	0.7*

* Threshold for IoU is not reported in [1]
/ Precision and recall score is not reported in [1]

The models fine-tuned on the IITP20 dataset, YOLOv11-m (IITP20) and YOLOv11-m (CCTV + IITP20), with initial weights from ImageNet and after adjustment to CCTV, respectively, achieve a mAP of 93%, which is significantly better than the previous results. In order to select a model that generalizes better for further experiments, we additionally tested them on images of real surveillance scenes collected from the Internet. It was shown that the combined training on different CCTV images contributed to the YOLOv11-m (CCTV + IITP20) model showing superiority in person and luggage detection on real surveillance images (Fig. 3, 4), making it our preferred choice for further development and application.

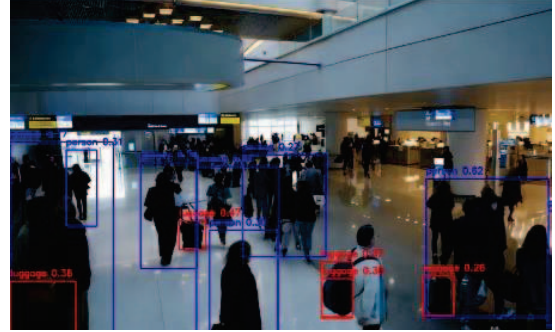


Fig. 3. Detection of YOLOv11-m (CCTV + IITP20) on unseen image



Fig. 4. Detection of YOLOv11-m (IITP20) on unseen image

IV. ABANDONED OBJECT DETECTION ALGORITHM WITH DYNAMIC ESTIMATION OF THE PERSON-BAGGAGE CONNECTION

In the following, we propose our new methodology for recognizing abandoned luggage within surveillance environments, which integrates real-time spatial depth information to dynamically estimate the person-baggage connection. The goal is to enable active adjustment of interaction zones based on distance from the camera to increase accuracy and reduce falsely positive abandoned luggage alarms.

A. Overview of Abandonment Detection Logic

Our abandonment detection logic combines elements of dynamic luggage-to-person distance measurement, temporal association mechanisms, and motion analysis. The determination of an item's abandonment status proceeds through the following stages:

1) Initial Assessment (Unattended Status Determination):

Prerequisite for Evaluation:

A fundamental condition in our methodology for any abandonment assessment is that the luggage must be stationary. If a luggage item is identified as moving, it is immediately removed from consideration as an abandoned object.

Proximity Check:

For objects confirmed to be stationary, the system then evaluates whether any currently associated individuals are present within their dynamically calculated distance radius, Fig. 5.

This step establishes the direct link between the person and the luggage, i.e. ownership of baggage.



Fig. 5. Mapping person and luggage based on distance within a given radius (yellow circle) in our abandoned luggage detection system

Classification:

If no association between person and luggage is detected within this designated zone, the luggage is classified as potentially unattended.

2) Transition to "Potential Abandonment" State:

Temporal Threshold:

If luggage remains stationary and unattended for a predefined period of time, it enters a "potential abandonment" state, Fig. 6. This initiates a temporal counter from the moment the potential unattended status is confirmed.

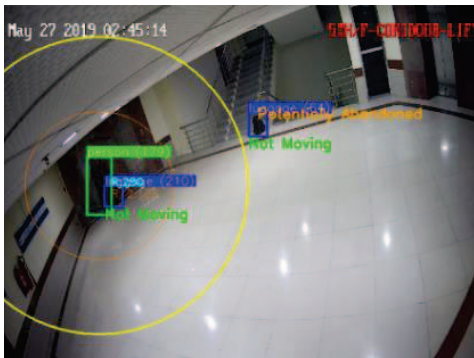


Fig. 6. Potentially abandoned state in our abandoned luggage detection system marked with "Potential Abandonment" text

Purpose:

This stage serves as an initial alert phase, flagging items that have remained isolated for a significant, yet not

definitively conclusive, period. It provides an early indication without triggering a confirmed alert.

3) Confirmation of "Definite Abandonment":

Expanded Spatial Verification:

To establish "definite abandonment," the system employs a distinct, larger definite abandonment radius. This radius, consistent with our spatial adaptation strategy, is also dynamically scaled based on depth (Section IV.B) and is typically a wider multiple of the effective proximity radius.

Objective:

The primary objective of this wider spatial evaluation is to ascertain the presence of any individual, including those not previously identified as an owner within a broader vicinity. This expanded check accommodates various real-world scenarios, such as a primary associate temporarily moving just beyond their immediate interaction zone or other individuals maintaining passive observation of the item.

Final Confirmation:

If there are no people in a wider radius around the baggage, and the baggage has spent a predefined time in the state of "potential abandonment", Fig. 7, starting from the moment of marking "potential abandonment", then the baggage is definitely declared "abandoned" (Fig. 7, left).



Fig. 7. Potential abandonment with timers (here 8 s) in our abandoned luggage detection system

B. Dynamic Radius Estimation in Person-baggage connection

A fundamental limitation in traditional object detection and tracking in video surveillance arises from perspective effects. A fixed pixel-based radius around an object in a two-dimensional image represents significantly different distances in the real world depending on the object's actual depth from the camera. This inherent distortion can lead to misinterpretations of distance, where the same pixel-based distance for objects far from the camera represents a much greater distance in the real world than for objects close to the camera.

To restore the depth information missing from the image plane, our system employs a depth perception model, specifically Apple Depth Pro [12].

Upon detecting a luggage item, its spatial coordinates are correlated with the corresponding estimation of depth information derived from the depth map, which provides a robust estimate of the luggage's true distance from the camera, Fig. 8. Based on this estimated depth, the system dynamically scales the effective distance radius for each

individual person, so the closer the luggage is to the camera (indicated by a smaller depth value), the larger its corresponding effective distance radius becomes. This accounts for the expanded visual field and greater area of interaction an object commands when viewed up close.

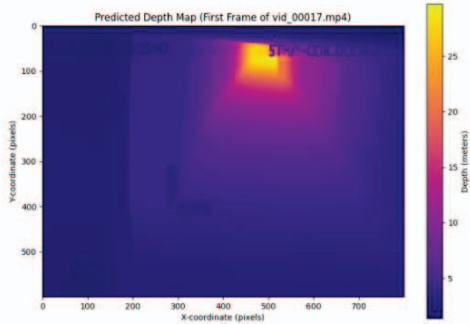


Fig. 8. Depth map of first frame of testing video

Conversely, luggage positioned farther away from the camera (larger depth value), appearing smaller in the frame, is assigned to a proportionally smaller effective radius, Fig. 9.



Fig. 9. Depth based radius size in our abandoned luggage detection system (On a nearby object, the radius is 250 and on objects further away, it is adjusted to 125)

The purpose of this dynamic adjustment is to normalize the perceived interaction area across varying depths, providing a more reliable and context-aware measure of an individual's general association with the luggage than any static pixel-based threshold could offer.

C. Refined Connection Establishment: The Inner Proximity Timer

Beyond assessing general distance, our system incorporates an "inner proximity" mechanism to establish and confirm intentional connections between individuals and luggage, Fig. 10. This component is intended for identifying new ownership, re-establishing a connection after a person has briefly moved away, or distinguishing a genuine interaction from a casual pass-by.

Upon entry of a person into this inner proximity zone around the stationary baggage, a timer is started for that specific person-object pairing. The primary purpose of this timer, along with the small area of the inner zone, is to determine actual, intentional interaction between the person and the baggage. If the person's continued presence within this small space lasts longer than a predetermined duration

for establishing a connection, a confirmed connection between that person and the baggage is formally registered.



Fig. 10. Close inner proximity radius (orange circle) in our abandoned luggage detection system

This timing condition filters out transient or accidental approaches to the baggage, ensuring that only sustained, intentional interactions are interpreted as a legitimate connection or re-establishment of a connection. A key functional advantage of this mechanism is its immediate ability to reset any pending states or actively abandon previously assigned baggage upon successful establishment of a new connection, thereby reflecting the object's re-attended status and mitigating false positive abandonment alerts.

D. Object Movement Analysis and Re-mapping Across Frames

The system continuously monitors all detected luggage items for any change in their spatial coordinates between successive frames. This involves comparing the current estimated position of each object with its position in the immediately preceding frame, or across a short sequence of recent frames, to identify displacement. Small, negligible displacements that might arise from sensor noise or minor jitter are filtered out using a predefined motion threshold. If an object's displacement exceeds this threshold over a given period, it is classified as moving.

The detection of movement has an immediate and decisive impact on the object's potential abandonment status. A core principle of our abandoned luggage detection algorithm is that if a luggage item is determined to be moving, it cannot, by definition, be considered abandoned. An actively moving object inherently implies a form of current interaction or transport, negating any state of being unattended and discarded. Therefore, any pending "potential abandonment" flags or active "definite abandonment" alerts associated with a moving item are immediately cleared.

Furthermore, unlike some approaches that might analyze the specific direction or velocity of a moving object, our framework posits that for abandonment detection, the mere presence of significant movement is sufficient to exclude an object from being considered abandoned. This step is employed for minimizing false positives, as it ensures that only genuine stationary and unattended items proceed through the abandonment assessment pipeline.

E. Object Re-mapping and Tracking Across Frames

To effectively monitor object movement and maintain continuity for abandonment detection, luggage items are continuously remapped or tracked across consecutive video

frames. In highly crowded environments, maintaining consistent object identities (IDs) using traditional long-term trackers can be challenging due to frequent occlusions, close proximities, and inter-object interactions. Such conditions often lead to ID switches or tracking failures [17], compromising the reliability of downstream abandonment detection. To mitigate these limitations and ensure robust object continuity, our system employs a re-mapping strategy. Rather than relying solely on a fixed tracker maintaining static IDs over extended periods, we integrate re-mapping upon significant movement changes of luggage and during each new object detection cycle. This adaptive re-mapping strategy thus assures that our system can maintain object continuity and perform accurate abandonment detection even when conventional long-term trackers struggle to preserve stable IDs in complex, crowded scenarios. When an object is successfully remapped to its identity from a previous frame, its trajectory and movement status are updated. This continuous tracking allows our system to:

- **Determine if an object has stopped:** By analyzing the object's movement status over time, the system can precisely identify when a previously moving object becomes stationary. This transition is important, as only then does the object become a candidate for abandonment.
- **Identify displacement:** Persistent re-mapping allows the system to calculate the cumulative distance an object has moved, which informs the movement detection described in the section above

In essence, the continuous tracking of objects provides the necessary temporal context for our abandonment detection framework. By accurately distinguishing between moving and stationary items and by persistently monitoring the state of each piece of luggage over time, the system can reliably determine when an object transitions from being actively managed to potentially be abandoned.

V. EXPERIMENTAL RESULTS

This section presents the performance evaluation of our proposed abandoned luggage system and discusses the specific characteristics of the dataset utilized for validation.

The primary dataset (IITP20) employed for our evaluation is derived from a collection of surveillance videos, which are broadly categorized into “Hard”, “Medium”, and “Easy” scenarios of abandonment [1]. It must be noted that the original dataset lacks precise, frame-level ground truth annotations for abandoned objects. Instead, videos are pre-split into “abandoned” and “not abandoned” folders based on these difficulty categories. This inherent characteristic of the dataset presents certain challenges for quantitative evaluation, as the exact moment and nature of abandonment are not granularly marked.

The dataset employed for our evaluation is a collection of 58 surveillance videos specifically designed to test abandoned object detection in various scenarios, categorized by complexity:

Easy Class: Comprises 19 videos of simple scenarios with few people and objects in full view. It differentiates from other datasets by including cases where objects are placed on chairs or high platforms, and these videos also feature radial distortion.

Medium Class: Consists of 20 videos that incorporate the complexities of the easy class, along with scenarios involving groups of people and instances of person re-entry.

Hard Class: This category includes 19 videos with complexities beyond those in the easy and medium classes. These cases feature hidden or partially visible objects, object exchanges, multiple owners for a single object, and single owners for multiple objects.

Furthermore, a review of the dataset revealed several instances where the video's categorization as “abandoned” does not align with our systems or the real-life meaning of abandonment, which requires an item to be both stationary and truly unattended without implied active association.

Specifically:

Group Separation Scenarios: Some videos labeled “abandoned” depict a group of people, with one individual leaving while the luggage remains with others (Figs 11, 12, 13). Our system does not consider such cases as true abandonment, as the luggage remains implicitly attended by the remaining individuals, even if the primary “owner” has departed.



Fig. 11. Group scenario (supervision by an acquainted individual)



Fig. 12. Luggage remains with a person sitting (not abandoned)



Fig. 13. Our system keeps both luggage items mapped to a person

Designated Storage Areas: The dataset includes examples of luggage being placed in designated storage areas (e.g., luggage closets or lockers, Fig. 14) and subsequently labeled

as “abandoned” Consistent with real-world security protocols, our system does not classify items placed in such designated areas as abandoned, as these are intended storage locations and do not imply a security risk.



Fig. 14. Luggage storage and “abandoned” luggage in them

Therefore, we redefined the ground truth in the dataset to include only scenes truly relevant to such systems. This step involved re-evaluating and, in some cases, reclassifying instances where original labels for “abandoned” or “not abandoned” luggage did not align with a real-world understanding of the term “abandoned”. This process ensured that the refined ground truth accurately reflected situations where luggage was genuinely unattended and posed a potential threat, thereby reclassifying some instances in both directions.

The abandoned baggage detection accuracy achieved by our proposed abandoned baggage algorithm based on the fine-tuned YOLOv11 model compared to the performance of the abandoned baggage model proposed in [1] is shown in Table 3.

Table 3. Accuracy of abandoned algorithms on IITP20 abandoned dataset

Category	Amin et. al [1] abandoned algorithm with ResNet	Proposed abandoned algorithm with YOLOv11 (CCTV+IITP20)
Hard	0.74	0.9
Medium	0.85	0.85
Easy	1	0.95
Total	0.86	0.9

The results demonstrate the robust performance of our abandoned object detection algorithm, which achieves an overall accuracy of 90%, especially when evaluated on difficult abandoned cases that are contextually relevant to surveillance systems in public places.

VI. LIMITATIONS AND FUTURE WORK

While our proposed YOLOv11-based algorithm with dynamic radius estimation demonstrates promising performance in detecting abandoned luggage, it is important to acknowledge certain limitations that present avenues for future research. One key challenge is that the model and its underlying tracker need to be fine-tuned for the specific environment in which they will be deployed. This is particularly critical in crowded areas, where the performance of object trackers can degrade. In such dynamic and complex scenes, trackers often struggle with identity changes (ID

switches), as people and luggage move in and out of the scene or are occluded. Furthermore, a significant challenge in real-world applications is the vast variety of luggage shapes, sizes, and colors. The model's performance may not be consistent over time when faced with a wide range of new and unencountered bag types. However, this issue can be mitigated through retraining on a more diverse dataset for each specific area, which would help the model learn and become more generalized.

Future work will focus on addressing these limitations by exploring more robust background modeling techniques and refining the dynamic radius estimation to handle more complex social interactions and crowded scenarios.

VII. CONCLUSION

In this paper, we introduced a novel and robust methodology for real-time abandoned object detection in complex surveillance environments. Addressing the limitations of traditional approaches, our framework presented dynamically scaled depth-based proximity radius, that significantly enhance accuracy of spatial interactions by adapting to perspective changes. Furthermore, we incorporated in abandonment algorithm a refined temporal connection mechanism, including a dedicated inner proximity timer, for establishing and re-establishing person-luggage associations, in order to minimize impact related to fleeting or accidental interactions.

Experimental results demonstrated the significant efficacy of our proposed system. Our fine-tuned YOLOv11 model achieved excellent person and luggage detection results of 93% mAP@50, and the abandoned baggage detection algorithm also proved successful with an accuracy of 90%. The achieved performances especially on complex and contextually relevant scenarios confirm the robustness of the system and the practical applicability of our approach in identifying real security threats, highlighting its potential even in the absence of ideal training and testing data.

Our experiment showed that a major challenge in the field of abandoned baggage detection is the lack of comprehensive datasets with precise annotations and specific timestamps for abandonment events. In addition, the used IITP20 dataset had inherent contextual inconsistencies in labeling, as some cases were labeled as abandonment even though they were not alarming in surveillance areas such as luggage placement in storage or transfer of luggage responsibility among group members. Inaccuracies in the ground truth data were corrected and in future work, we plan to deal with generating a set of data from public monitored areas that will include different cases of luggage abandonment.

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